K-Means Algorithm Activity

DWDM CSA-1614

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Aim

To apply the K-Means Clustering Algorithm to different datasets, calculate Euclidean distances, assign data points to clusters, update centroids, and identify the final clusters after convergence.

General Formulae

Since all three problems contain one-dimensional data, the Euclidean distance between a data point x and a centroid C is calculated as:

$$d(x, C) = | x - C |$$

The centroid of each cluster is updated as the mean of the data points belonging to it:

$$C_i = (\text{Sum of data points in cluster } i) / (\text{Number of data points in cluster } i)$$

Note: In case of equal distances, the data point is assigned to the cluster with the smaller cluster number. The algorithm stops when the cluster assignments and centroids remain unchanged.

Algorithm Steps

1. Choose the number of clusters K and select K initial centroids.
2. Calculate the distance of every data point from each centroid.
3. Assign each data point to the cluster whose centroid is nearest (ties go to the smaller cluster number).
4. Recalculate each centroid as the mean of the data points assigned to its cluster.
5. Repeat Steps 2 to 4 until the cluster assignments and centroids no longer change.

Problem 1 – K = 3

Given Dataset: D = {2, 4, 6, 8, 10, 12, 14, 16, 18}

Initial Centroids: $C_1 = 2$, $C_2 = 8$, $C_3 = 14$

Iteration 1: Calculate Distances and Assign Clusters

The distance of each data point from all the initial centroids is calculated, and each point is assigned to the nearest centroid.

Table 1: Distance calculation and cluster assignment – Problem 1, Iteration 1

Data Point	$C_1 = 2$	$C_2 = 8$	$C_3 = 14$	Assigned Cluster
2	0	6	12	C_1
4	2	4	10	C_1
6	4	2	8	C_2
8	6	0	6	C_2
10	8	2	4	C_2
12	10	4	2	C_3
14	12	6	0	C_3
16	14	8	2	C_3
18	16	10	4	C_3

Cluster Formation and New Centroids

Table 2: Cluster formation and centroid update – Problem 1, Iteration 1

Cluster	Data Points	Centroid Calculation	New Centroid
C_1	{2, 4}	(2 + 4) / 2	3
C_2	{6, 8, 10}	(6 + 8 + 10) / 3	8
C_3	{12, 14, 16, 18}	(12 + 14 + 16 + 18) / 4	15

Updated centroids: $C_1 = 3$, $C_2 = 8$, $C_3 = 15$

Iteration 2: Recalculate Distances

Using the updated centroids (3, 8, 15), the distances are calculated again.

Table 3: Distance calculation and cluster assignment – Problem 1, Iteration 2

Data Point	$C_1 = 3$	$C_2 = 8$	$C_3 = 15$	Assigned Cluster
2	1	6	13	C_1
4	1	4	11	C_1
6	3	2	9	C_2
8	5	0	7	C_2
10	7	2	5	C_2
12	9	4	3	C_3
14	11	6	1	C_3
16	13	8	1	C_3
18	15	10	3	C_3

Convergence Check

The cluster assignments remain unchanged:

- $C_1 = \{2, 4\}$
- $C_2 = \{6, 8, 10\}$
- $C_3 = \{12, 14, 16, 18\}$

The recalculated centroids are still 3, 8, 15. Hence, the algorithm has converged.

Final Answer – Problem 1

Table 4: Final clusters – Problem 1

Cluster	Final Data Points	Final Centroid
C_1	2, 4	3
C_2	6, 8, 10	8
C_3	12, 14, 16, 18	15

Result: The K-Means algorithm successfully formed three clusters, and the centroids remained unchanged after the second iteration.

Problem 2 – K = 4

Given Dataset: D = {5, 10, 15, 20, 25, 30, 35, 40, 45, 50}

Initial Centroids: $C_1 = 5$, $C_2 = 15$, $C_3 = 30$, $C_4 = 45$

Iteration 1: Calculate Distances and Assign Clusters

The distance of each data point from all the initial centroids is calculated, and each point is assigned to the nearest centroid.

Table 5: Distance calculation and cluster assignment – Problem 2, Iteration 1

Data Point	$C_1 = 5$	$C_2 = 15$	$C_3 = 30$	$C_4 = 45$	Assigned Cluster
5	0	10	25	40	C_1
10	5	5	20	35	C_1
15	10	0	15	30	C_2
20	15	5	10	25	C_2
25	20	10	5	20	C_3
30	25	15	0	15	C_3
35	30	20	5	10	C_3
40	35	25	10	5	C_4
45	40	30	15	0	C_4
50	45	35	20	5	C_4

Note: For data point 10, the distances to C_1 and C_2 are equal (5). It is assigned to C_1 according to the tie-breaking rule.

Cluster Formation and New Centroids

Table 6: Cluster formation and centroid update – Problem 2, Iteration 1

Cluster	Data Points	Centroid Calculation	New Centroid
C_1	{5, 10}	(5 + 10) / 2	7.5
C_2	{15, 20}	(15 + 20) / 2	17.5
C_3	{25, 30, 35}	(25 + 30 + 35) / 3	30
C_4	{40, 45, 50}	(40 + 45 + 50) / 3	45

Updated centroids: $C_1 = 7.5$, $C_2 = 17.5$, $C_3 = 30$, $C_4 = 45$

Iteration 2: Recalculate Distances

Using the updated centroids (7.5, 17.5, 30, 45), the distances are calculated again.

Table 7: Distance calculation and cluster assignment – Problem 2, Iteration 2

Data Point	$C_1 = 7.5$	$C_2 = 17.5$	$C_3 = 30$	$C_4 = 45$	Assigned Cluster
5	2.5	12.5	25	40	C_1
10	2.5	7.5	20	35	C_1
15	7.5	2.5	15	30	C_2
20	12.5	2.5	10	25	C_2
25	17.5	7.5	5	20	C_3
30	22.5	12.5	0	15	C_3
35	27.5	17.5	5	10	C_3
40	32.5	22.5	10	5	C_4
45	37.5	27.5	15	0	C_4
50	42.5	32.5	20	5	C_4

Convergence Check

The cluster assignments remain unchanged:

- $C_1 = \{5, 10\}$
- $C_2 = \{15, 20\}$
- $C_3 = \{25, 30, 35\}$
- $C_4 = \{40, 45, 50\}$

The recalculated centroids are still 7.5, 17.5, 30, 45. Hence, the algorithm has converged.

Final Answer – Problem 2

Table 8: Final clusters – Problem 2

Cluster	Final Data Points	Final Centroid
C_1	5, 10	7.5
C_2	15, 20	17.5
C_3	25, 30, 35	30
C_4	40, 45, 50	45

Result: The K-Means algorithm successfully formed four clusters, and the centroids remained unchanged after the second iteration.

Problem 3 – K = 5

Given Dataset: D = {3, 6, 9, 15, 18, 24, 27, 33, 36, 42}

Initial Centroids: $C_1 = 3$, $C_2 = 9$, $C_3 = 18$, $C_4 = 27$, $C_5 = 36$

Iteration 1: Calculate Distances and Assign Clusters

The distance of each data point from all the initial centroids is calculated, and each point is assigned to the nearest centroid.

Table 9: Distance calculation and cluster assignment – Problem 3, Iteration 1

Data Point	$C_1 = 3$	$C_2 = 9$	$C_3 = 18$	$C_4 = 27$	$C_5 = 36$	Assigned Cluster
3	0	6	15	24	33	C_1
6	3	3	12	21	30	C_1
9	6	0	9	18	27	C_2
15	12	6	3	12	21	C_3
18	15	9	0	9	18	C_3
24	21	15	6	3	12	C_4
27	24	18	9	0	9	C_4
33	30	24	15	6	3	C_5
36	33	27	18	9	0	C_5
42	39	33	24	15	6	C_5

Note: For data point 6, the distances to C_1 and C_2 are equal (3). It is assigned to C_1 using the tie-breaking rule.

Cluster Formation and New Centroids

Table 10: Cluster formation and centroid update – Problem 3, Iteration 1

Cluster	Data Points	Centroid Calculation	New Centroid
C_1	{3, 6}	(3 + 6) / 2	4.5
C_2	{9}	9 / 1	9
C_3	{15, 18}	(15 + 18) / 2	16.5
C_4	{24, 27}	(24 + 27) / 2	25.5
C_5	{33, 36, 42}	(33 + 36 + 42) / 3	37

Updated centroids: $C_1 = 4.5$, $C_2 = 9$, $C_3 = 16.5$, $C_4 = 25.5$, $C_5 = 37$

Iteration 2: Recalculate Distances

Using the updated centroids (4.5, 9, 16.5, 25.5, 37), the distances are calculated again.

Table 11: Distance calculation and cluster assignment – Problem 3, Iteration 2

Data Point	$C_1 = 4.5$	$C_2 = 9$	$C_3 = 16.5$	$C_4 = 25.5$	$C_5 = 37$	Assigned Cluster
3	1.5	6	13.5	22.5	34	C_1
6	1.5	3	10.5	19.5	31	C_1
9	4.5	0	7.5	16.5	28	C_2
15	10.5	6	1.5	10.5	22	C_3
18	13.5	9	1.5	7.5	19	C_3
24	19.5	15	7.5	1.5	13	C_4
27	22.5	18	10.5	1.5	10	C_4
33	28.5	24	16.5	7.5	4	C_5
36	31.5	27	19.5	10.5	1	C_5
42	37.5	33	25.5	16.5	5	C_5

Convergence Check

The cluster assignments remain unchanged:

- $C_1 = \{3, 6\}$
- $C_2 = \{9\}$
- $C_3 = \{15, 18\}$
- $C_4 = \{24, 27\}$
- $C_5 = \{33, 36, 42\}$

The recalculated centroids are still 4.5, 9, 16.5, 25.5, 37. Hence, the algorithm has converged.

Final Answer – Problem 3

Table 12: Final clusters – Problem 3

Cluster	Final Data Points	Final Centroid
C_1	3, 6	4.5
C_2	9	9
C_3	15, 18	16.5
C_4	24, 27	25.5
C_5	33, 36, 42	37

Result: The K-Means algorithm successfully formed five clusters, and the centroids remained unchanged after the second iteration.

Overall Conclusion

The K-Means Clustering Algorithm was successfully applied to all three numerical problems.

Table 13: Summary of results

Problem	Number of Clusters (K)	Final Centroids	Iterations
Problem 1	3	3, 8, 15	2
Problem 2	4	7.5, 17.5, 30, 45	2
Problem 3	5	4.5, 9, 16.5, 25.5, 37	2

The algorithm grouped the data points according to their nearest centroids and repeatedly updated the centroid values until convergence was achieved.

Final Observation: All three problems converged after the first centroid update, with the second iteration confirming that the cluster assignments and centroid values remained unchanged.